



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

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Department of Mechatronics Engineering 23MT507- Prototype Lab

Third Review

Date : 28/10/2025



DESIGN AND FABRICATION OF AUTOMATIC WASTE SEGREGATION ROVER SYSTEM

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ASSISTANT PROFESSOR - MCT

OBJECTIVE

- To design and develop an IoT-based smart garbage bin (Garbot) that enables real-time monitoring of waste levels and environmental conditions.
- To automate the waste collection process by generating timely alerts, thereby improving operational efficiency and reducing manual intervention.

ABSTRACT

An IoT-Based Smart Sustainable waste Management System

- Garbot is an IoT-enabled smart bin that monitors waste levels and ensures efficient, eco-friendly collection.
- This project introduces Garbot, a smart waste bin that automates waste tracking and supports sustainable cities.
- Garbot leverages IoT to reduce manual waste monitoring and promote cleaner, smarter urban environments.

INTRODUCTION

- **Smart Segregated Bins:** The rover includes separate compartments for wet, plastic, paper, and metal waste to enable proper source segregation.
- **Real-Time Fill Monitoring:** Ultrasonic sensors track bin fill levels and send alerts via a mobile app when any bin nears capacity.
- **Cover & Liner Detection:** Integrated sensors detect the status of the bin covers and liners, ensuring hygienic waste disposal by notifying when replacement is needed.
- **Remote-Controlled Mobility:** The rover can be navigated using a mobile app, allowing it to move towards users for convenient waste disposal.
- **Eco-Friendly Power System:** Operates on a rechargeable Battery power supply, making it sustainable and suitable for both home and public use.

LITERATURE REVIEW

S.NO	TITLE & AUTHORS	YEAR OF PUBLICATION	INFERENCE
1	Sustainable Smart Trash Bin Based Waste Segregation and Collection System Abhishek Singh,A. Banerjee,S Chatterjee,H.K. Das,N. K. Das,Sirshendu Saha,Hari D. Mandal	SEP 2024	- IoT-enabled smart bin with MQ137, TGS2611, and metal sensors. - Sorts biodegradable, plastic, metal, glass, and mixed waste at the source. - Improves recycling efficiency and reduces landfill burden.
2	Sort-O-Matic: An efficient Waste Segregation System for Sustainable Waste Management Rubina Dutta,Jeetasha Jeetasha,Harsimran Jit Kaur,Deepti Prit Kaur,Priyanka Malhotra	OCT 2023	- Uses metal proximity, moisture, IR, and ultrasonic sensors. - Reduces human contact with hazardous waste. - Enhances separation efficiency for different waste streams.
3	CNN Based Smart Waste Segregation and Collection System V. Lavanya,G. Abikash Shivam,S. Shanmugasundaram,R. Deepalakshmi	MAR 2023	- Uses YOLOv8 and CNN for real-time waste image classification. - Categorizes into biodegradable, recyclable, and non-recyclable. - Potentially scalable to urban waste management networks.

LITERATURE REVIEW

S.NO	TITLE & AUTHORS	YEAR OF PUBLICATION	INFERENCE
4	IOT based Waste Segregation Using Deep Learning Atharv Dahale	APR 2022	- Combines IoT fill-level sensors with CNN waste classification. - Supports smart scheduling of waste collection trucks. - Reduces overflow and improves recycling throughput.
5	Sustainable Waste Segregation, Sorting and Collection Protocols in Rural Touristic Areas Bhupinder Singh,Christian Kaunert,Pankaj Kumar Tyagi	NOV 2021	- Industrial IoT-based waste sorting for rural tourism locations. - Reduces waste leakage into natural landscapes. - Focuses on low-power, autonomous operation.
6	Development of an Automatic Waste Segregator and Monitoring System Kale & Patil	APR 2018	- Low-cost microcontroller-based system. - Detects and separates metal, wet, and dry waste using IR and moisture sensors. - Includes GSM-based bin monitoring for timely collection.

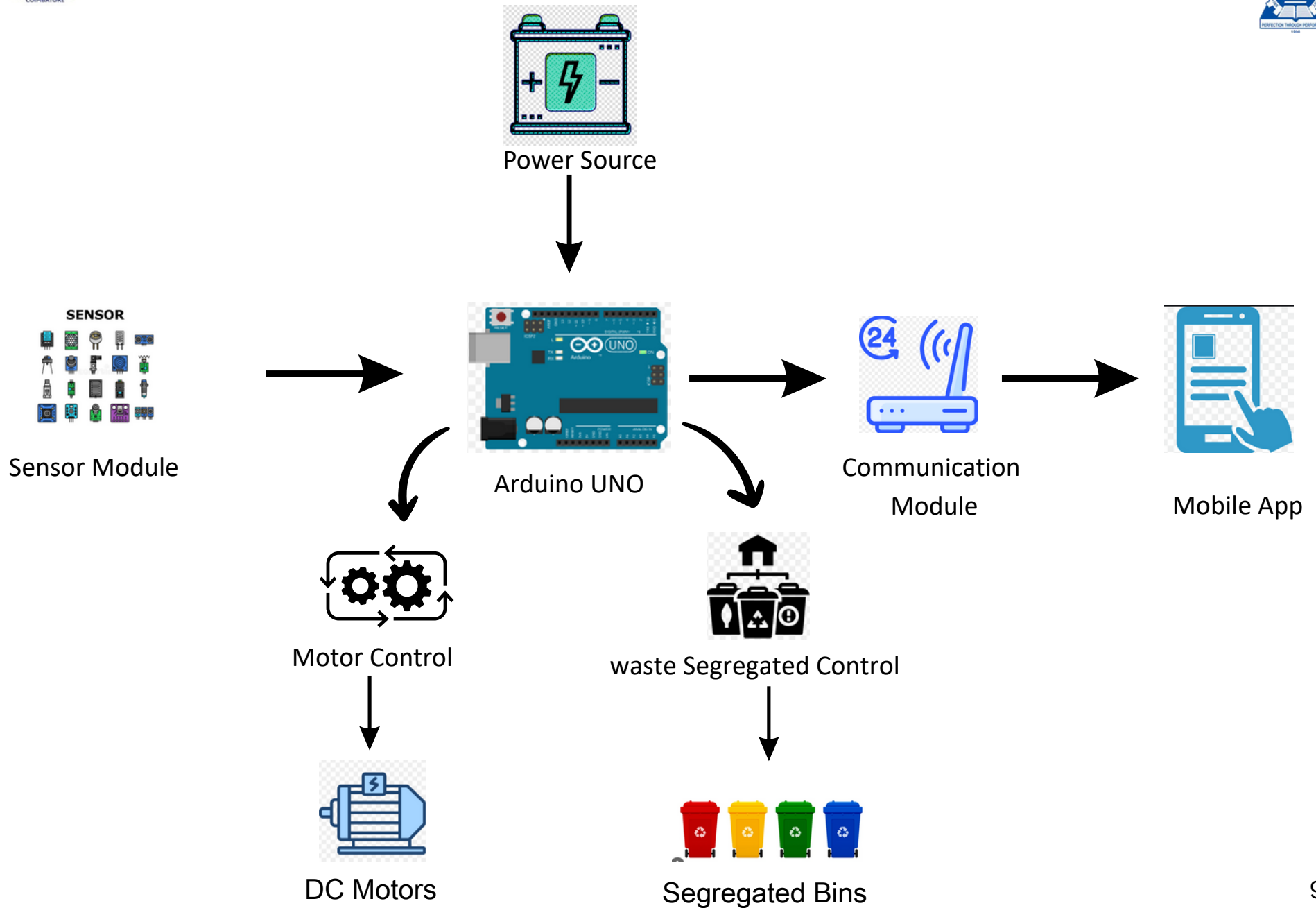
LITERATURE REVIEW

S.NO	TITLE & AUTHORS	YEAR OF PUBLICATION	INFERENCE
7	Design and Development of an Intelligent Waste Segregator for Smart Cities Rathod & Shukla Singh	JUNE 2018	• Targets urban applications with metal, wet, and dry waste segregation. • Arduino-based control with real-time alerts to collection units. • Emphasis on modular, scalable design for city-wide deployment.

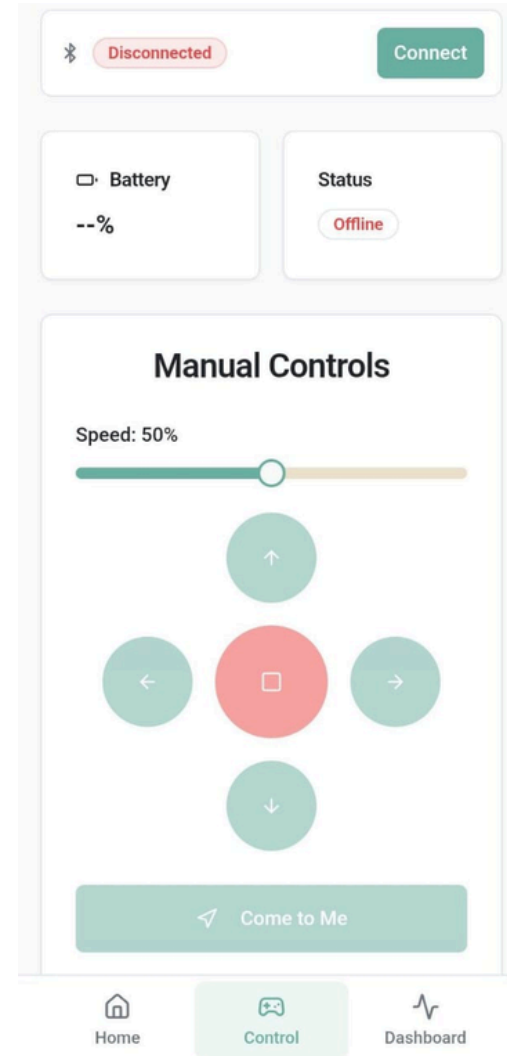
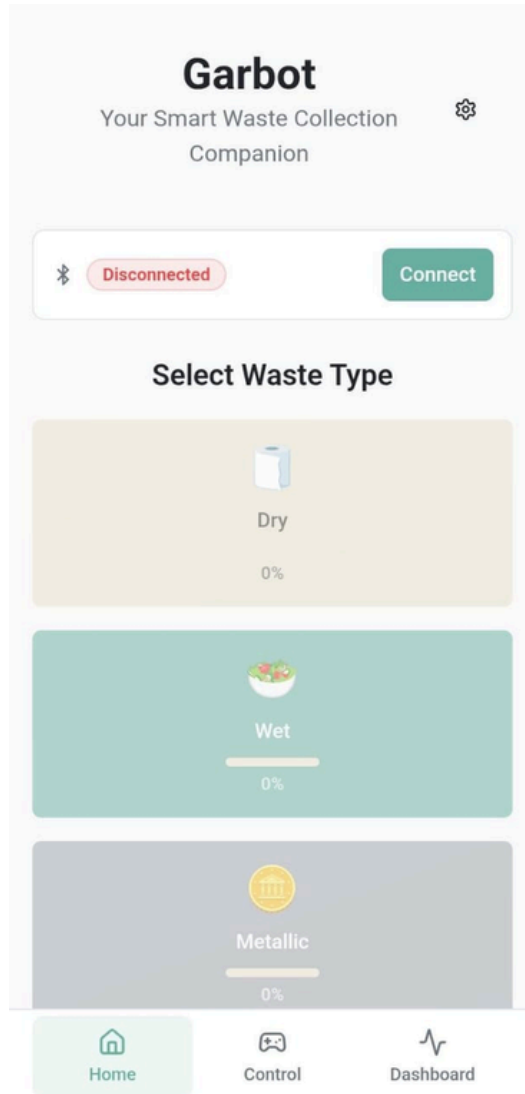
BILL OF MATERIALS

S.No	Parts	Quantity	Quantity
1	ARDUINO UNO REV3 SMD	1	600
2	ULTRASONIC SENSOR	1	100
3	Rain sensor	1	100
4	PCB Board (15*10)	1	600
5	5V SERVO Motor	2	500
6	ACRYLIC SHEET	1	250
7	JUMPER WIRES(M-F,F- F,M-M)	200	800
8	INDUCTIVE PROXIMITY SENSOR	1	350
9	ROCKER Switch	2	100
10	Battery(5000mAH)	1	1000
11	ESP32 MICROCONTROLLER	1	300
12	Gear Motor	4	500
13	L298N Motor Driver	2	300
14	PVC PIPE (15 CM)	1	20
		Total	5520

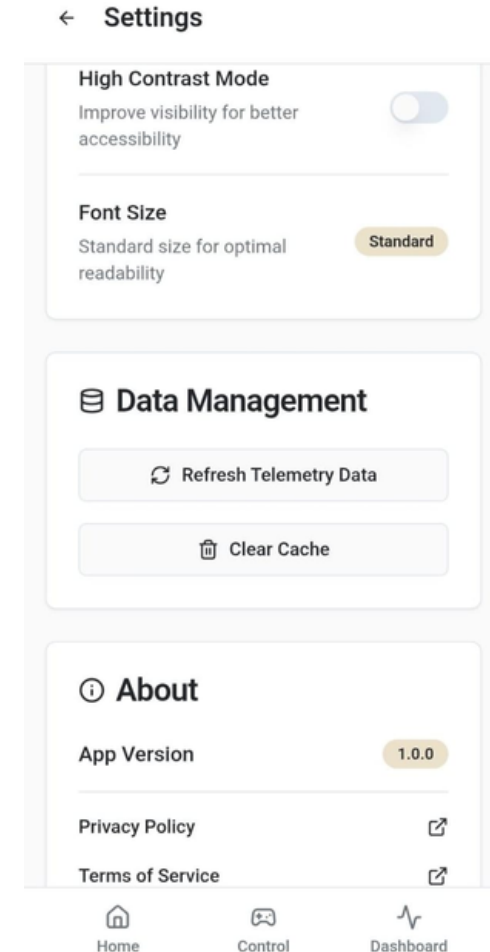
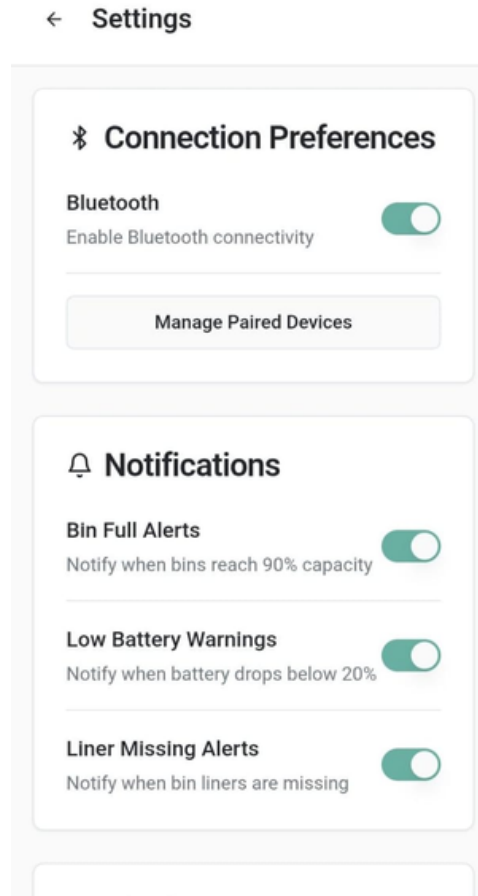
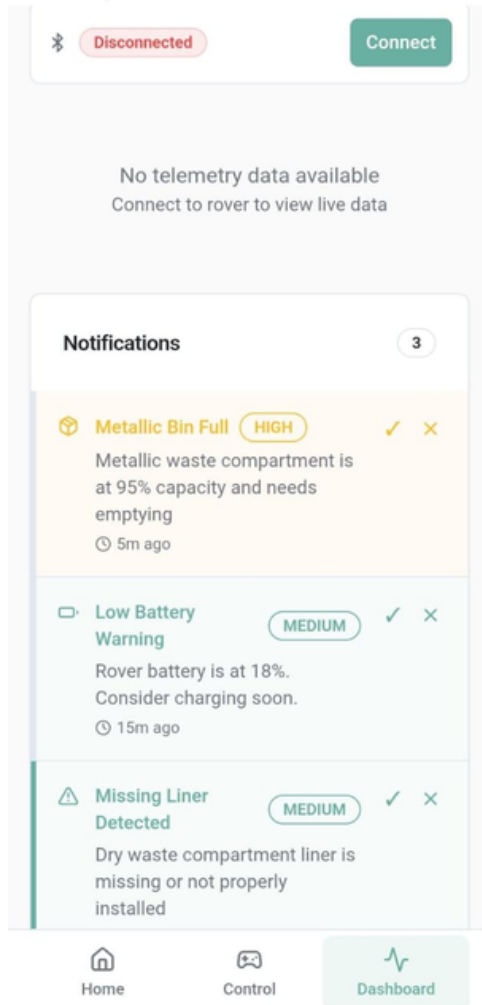
Block Diagram



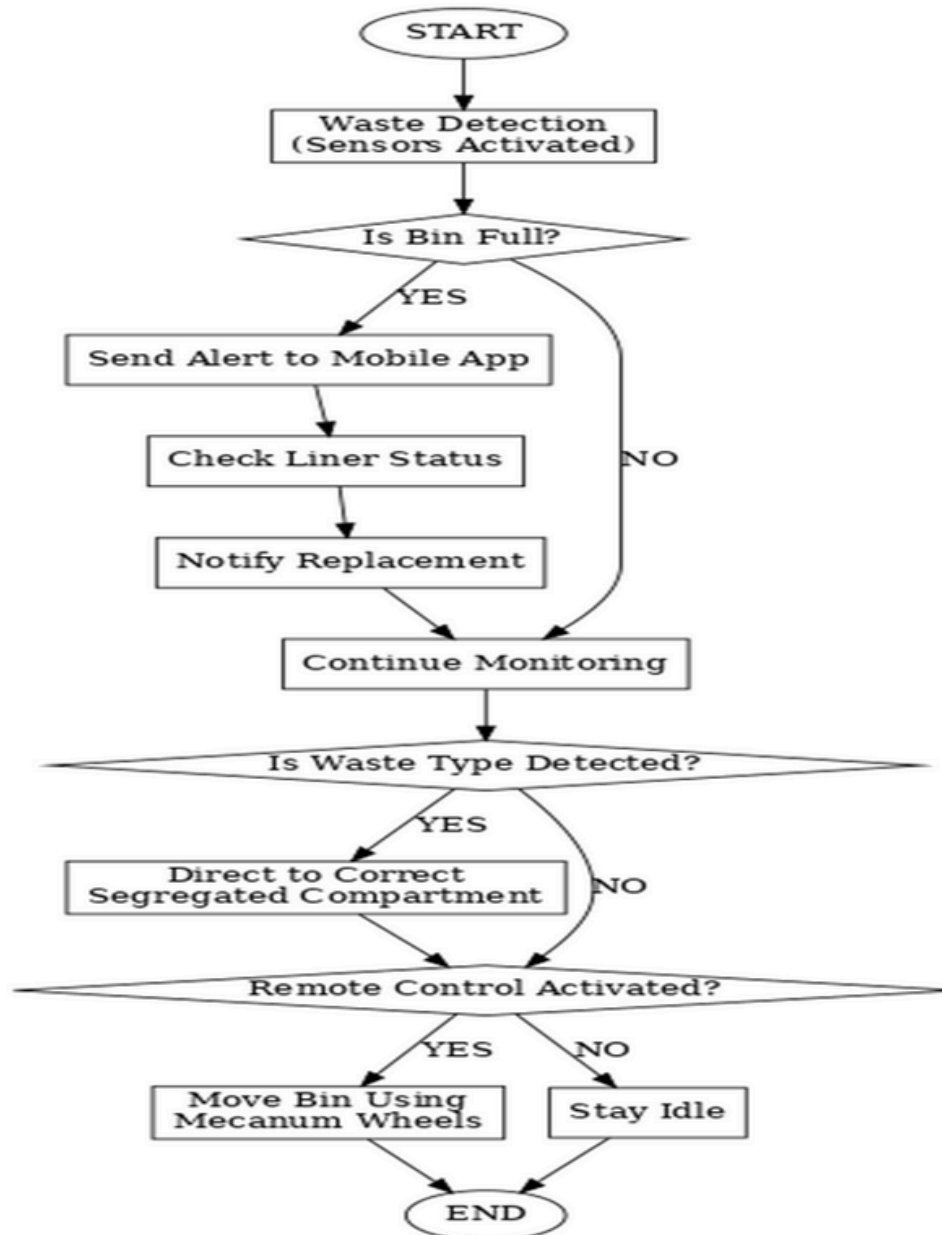
APP user Interface



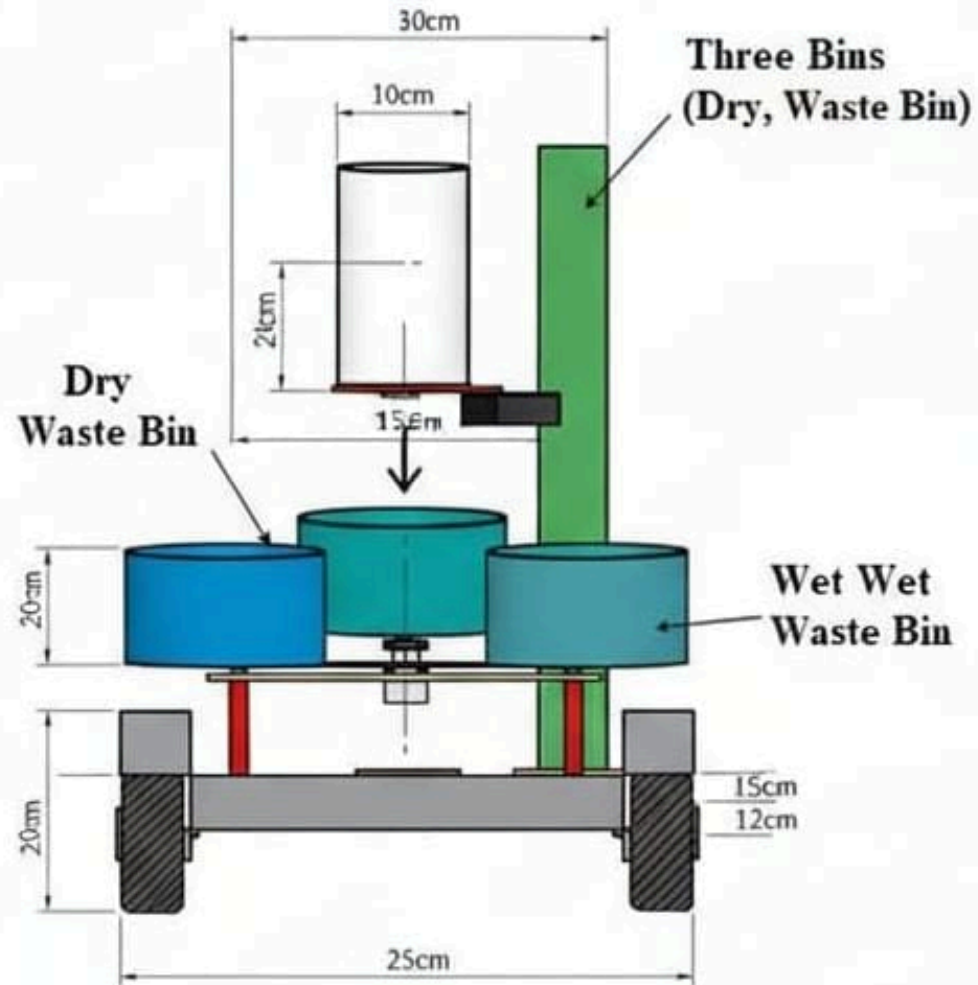
APP user Interface



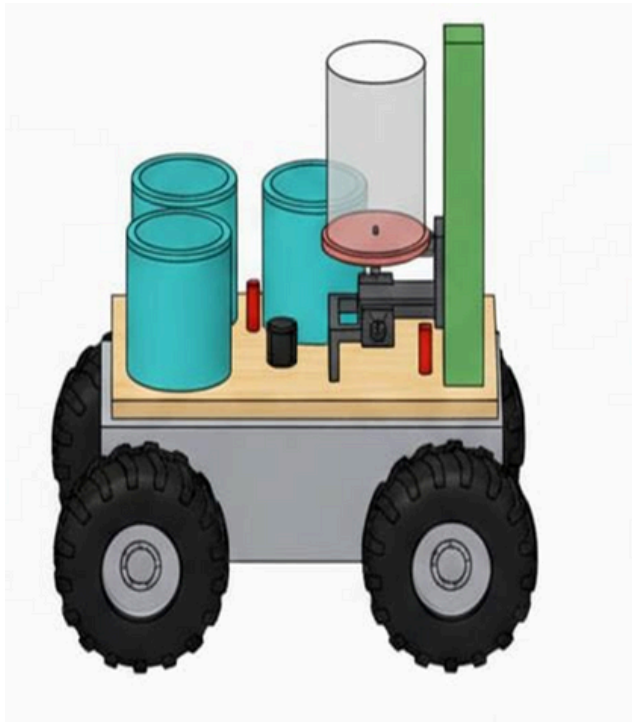
FLOWCHART



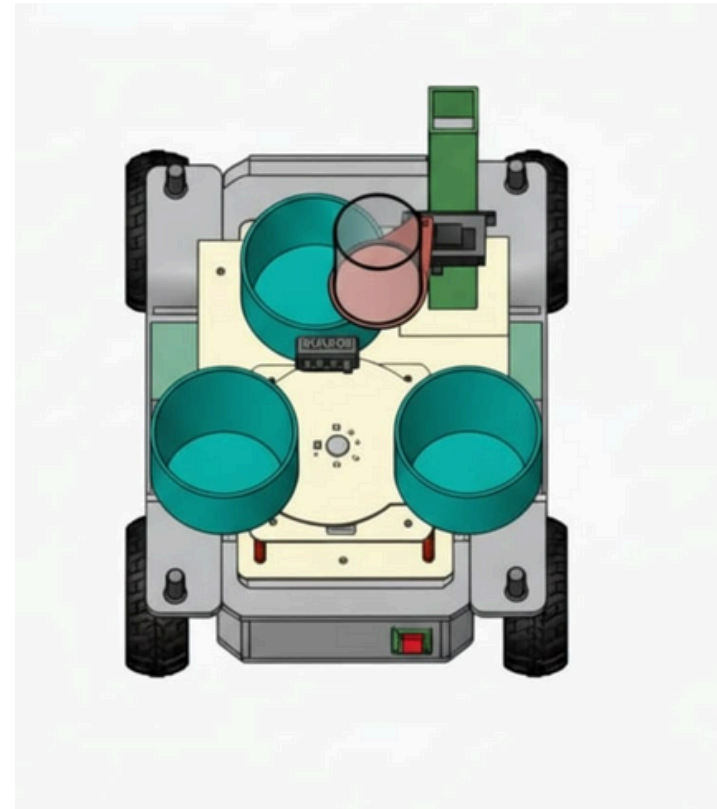
2D MODEL



3D MODEL

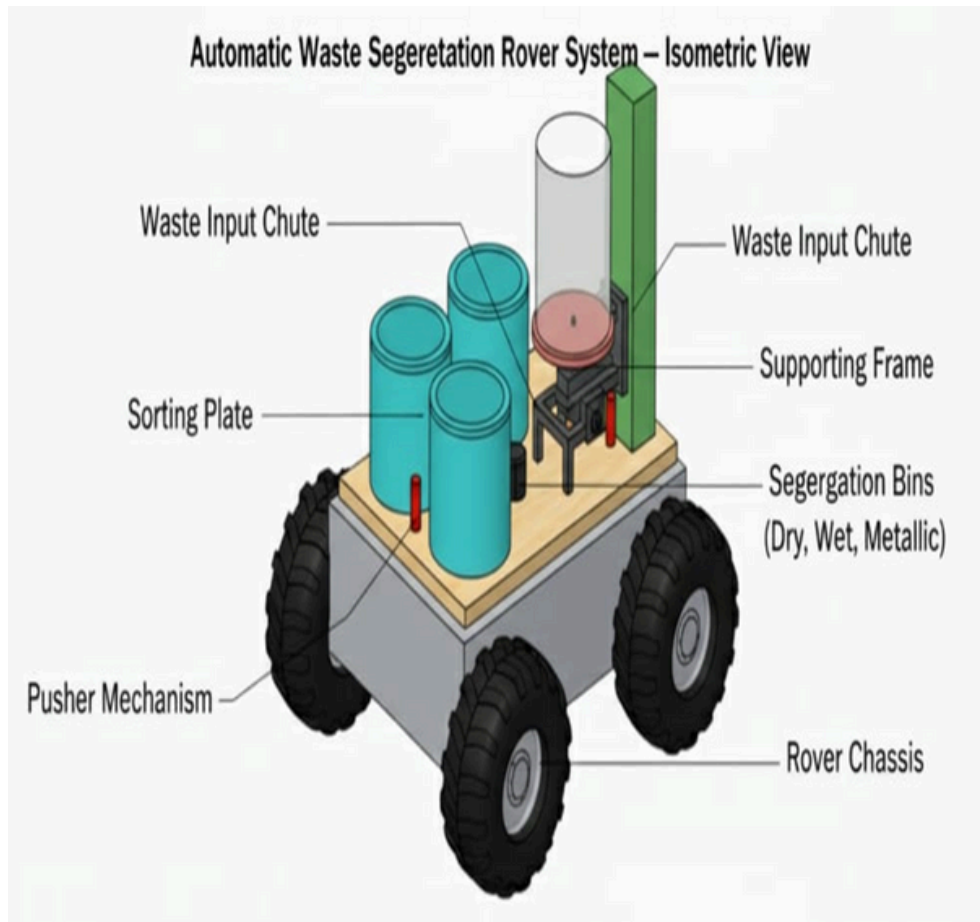


SIDE VIEW

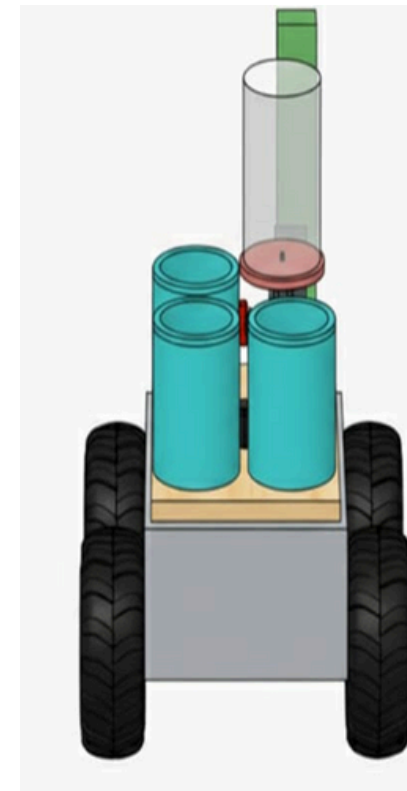


TOP VIEW

3D MODEL



ISOTROPIC VIEW



FRONT VIEW

DESIGN CALCULATION:

TOTAL WEIGHT = 4 Kg

Diameter Calculation:

Total no. of wheels, $n=4$

Weight exerted per wheel, $W = w/n$

$= 4/4$

$= 1\text{Kg per wheel}$

Power Required:

$P = w \times T$ Here,

$w = \text{Angular Velocity}$

$w = v/R$ $w = 1.39/0.03 = 46.33\text{rad/s}$

$P = 46.33 \times 0.2943 = 13.63\text{Watts (for 4 wheels)}$.

Power required for one Wheel $P = 13.63/4$

$P = 3.4\text{Watts}$

So for the required calculations SPG30E-60K Geared Motor is Chosen.

Torque Required:

$T = F_f \times r$ Here,

$F_f = w \times g$ $w = 1\text{Kg}$ (for 1 wheel)

$g = 9.81 \text{ m/s}^2$ (acceleration due to gravity)

$F_f = 1 \times 9.81\text{N}$

$= 9.81\text{N}$

$T = 9.81 \times 0.03$ (Wheel radius in m)

$T = 0.2943\text{Nm}$

$T = 3\text{Kgf.cm}$

This Torque is for One Wheel.

So, for Four Wheels,

$T = 4 \times 0.2943$

$T = 1.17\text{Nm}$ or 11.9Kgf.cm

Bending Moment for Chasis:

Weight, $W = 4\text{Kg}$

Length, $L = 26\text{cm}$

$= 260\text{mm}$

Breadth, $b = 21\text{cm}$

$= 210\text{mm}$

Uniformly Distributed Load:

Moment of Inertia:

$$I = (1/12)bxh^2$$

$$= (1/12)260 \times 210^3$$

$$= 38281.19\text{mm}^4$$

Determining Load Distribution:

$w = \text{Total load} / \text{Length}$

$$= 4/260$$

$$= 0.0153\text{Kg/mm}$$

DESIGN CALCULATION:



Square:

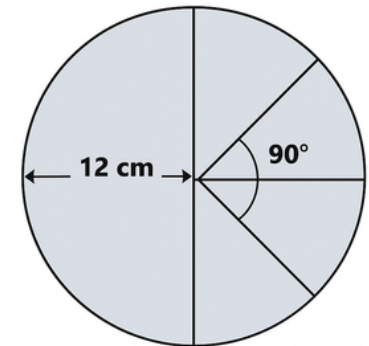
Length, $L=30\text{cm}$

Breadth, $B=30\text{mm}$

Circle:

Radius (r) = 12 cm

Radius (r) = 12 cm



1. Angle of Each Segment (Sector):

A full circle is 360° , so each segment (sector) will have:

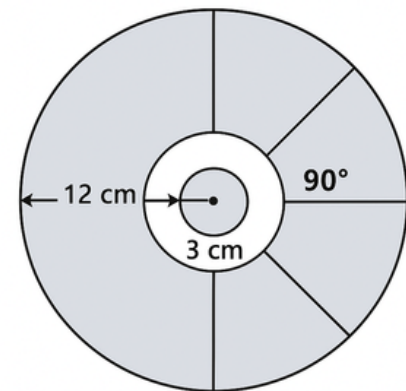
$$\text{Angle} = 360/4$$

$$= 90^\circ$$

2. Area of Full Circle:

$$A = \pi r^2 = \pi \times 12^2 = 144\pi$$

$$= 452.39 \text{ cm}^2$$



DESIGN CALCULATION:



3. Area of Each Segment (Quarter-Circle):

$$A_{\text{segment}} = \frac{90}{360} \times \pi r^2$$

$$= \frac{1}{4} \times \pi \times 12^2$$

$$= \frac{1}{4} \times 144\pi$$

$$= 36\pi$$

$$\approx \mathbf{113.10 \text{ cm}^2}$$

4. Arc Length of Each Segment:

Arc length formula:

$$L = \frac{\theta}{360} \times 2\pi r$$

$$= \frac{90}{360} \times 2\pi \times 12$$

$$= \frac{1}{4} \times 24\pi = 6\pi$$

$$\approx \mathbf{18.85 \text{ cm}}$$

5. Chord Length of Each Segment:

This is the straight line connecting the two ends of the arc (forms a triangle with the center). In a 90° segment:

$$\text{Chord length} = \sqrt{2} \times r = \sqrt{2} \times 12$$

$$\approx \mathbf{16.97 \text{ cm}}$$

DESIGN CALCULATION:

ACRYLIC SHEET BASE:

Size: 30 cm* 30 cm

Thickness: 4mm(0.4 cm)

$$\begin{aligned}\text{Volume} &= 30*30*0.4 \\ &= \mathbf{360 \text{ cm}^3}\end{aligned}$$

Density of Acrylic = 1.18g/cm^3

$$\begin{aligned}\text{Mass} &= 360*1.18 = 425\text{g} \\ &= \mathbf{0.43\text{Kg}}\end{aligned}$$

therefore, Base Weight = 0.43 kg

Load check :

Safe for loads <3kg 9since design limit ~5kg).

Rover Base: (20*20cm)

DESIGN CALCULATION:

Total Weight

Mass of the battery = 0.135 kg

$$\begin{aligned}\text{Weight of the battery} &= 0.135 * 9.8 \\ &= \mathbf{1.323N}\end{aligned}$$

Mass of Gear motor = 0.5 Kg

$$\begin{aligned}\text{Weight of Gear motor} &= 0.5 * 9.8 \\ &= \mathbf{4.9 N}\end{aligned}$$

$$\begin{aligned}\text{Weight of Four gear motors} &= 4 * 4.9 \\ &= \mathbf{19.6 Kg}\end{aligned}$$

DESIGN CALCULATION:

CURRENT AND BATTERY CALCULATION:

CURRENT CALCULATION:

Operating voltage = 12 v

Power = Voltage * Current

Current (I) = Power/Voltage

$$= 30.72 / 12$$

$$= 2.56 \text{ A}$$

BATTERY CALCULATION:

Battery capacity(Ah) = Current * Operating time of battery

Operating Time of battery = 3 Hours

Battery capacity(Ah) = Current * Operating Time = 2.56 * 3

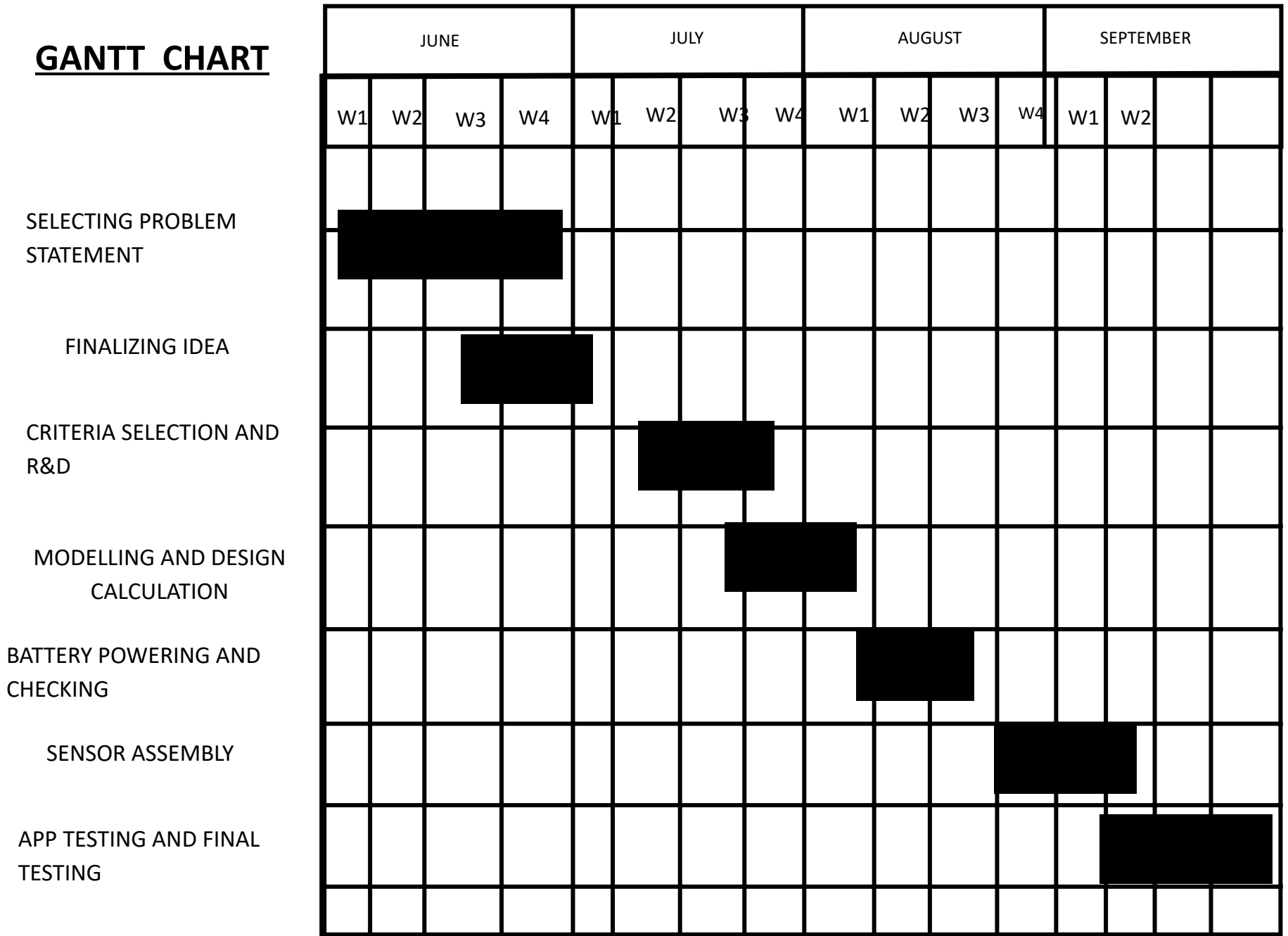
$$= 7.8 \text{ Ah}$$

Thus, the design calculations for our project were done. These calculations were done keeping in mind the dimensional restrictions for this project, along with economic views. The values are well within the standard limit and the design is safe and accurate for implementation

PROPOSED OUTCOME

- Efficient IoT-enabled waste monitoring with reduced overflow and optimized collection routes.
- Autonomous robotic system ensuring safer, cleaner, and sustainable waste handling.
- Data-driven insights for smart city waste management and resource optimization.

GANTT CHART



REFERENCES

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- 3) IEEE Smart Cities Initiative. (2023). Trends in IoT and Waste Management Solutions – IEEE Smart Cities White Paper.
- 4) Circuit Digest. (2023). Smart Dustbin Using Arduino and Ultrasonic Sensor – <https://circuitdigest.com>.
- 5) Reddy, K. et al. (2020). IoT Based Waste Segregator for Smart Cities – International Journal of Recent Technology and Engineering (IJRTE).
- 6) Thakker, S., & Narayanamoorthi, R. (2019). Smart and Wireless Waste Management System – IEEE Xplore.

THANK
YOU